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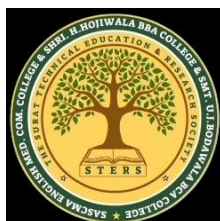
SASCMA English Medium Commerce College (B.Com. & M.Com.) &

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A PROJECT REPORT ON ONLINE SHOES WEBSITE

(StepUp Footwear)

**SUBMITTED AS A PARTIAL REQUIREMENT
FOR
BACHELOR OF COMPUTER APPLICATION
(SEMESTER V)**

YEAR 2026-27

GROUP NO.: 129

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1 INTRODUCTION

The footwear market is growing rapidly with diverse choices ranging from everyday casual sneakers to high-performance running and sports shoes. In conventional shoe stores, customers have to travel physically to the retail shop, spend time searching through crowded racks, and often face disappointment due to non-availability of their required shoe size or preferred brand model. Moreover, managing customer inquiries, physical stock registers, order receipts, and paper billing creates heavy manual work for retail store owners.

To solve these real-world problems, this project titled **"Online Shoes Website" (StepUp Footwear)** has been developed as an e-commerce platform for BCA Semester 5 (AWD). It offers a centralized web portal where customers can browse shoes from leading brands like Nike, Adidas, Puma, Converse, Skechers, Woodland, and Clarks, view detailed specifications, select shoe sizes, manage their cart and wishlist, and place orders with online payment. An independent administrator panel allows store staff to manage product inventory, categories, brand listings, customer orders, and customer reviews efficiently.

1.1 Project Profile

The general overview and key parameters of the project are outlined in the table below:

Project Title	Online Shoes Website (StepUp Footwear)
Subject	Advanced Web Development (AWD) — BCA Semester 5
Academic Year	2026 - 2027
Project Type	Full-Stack E-Commerce Web Application
Front-End Technologies	React.js (Vite), React Router DOM v6, Context API, CSS3
Back-End Technologies	Node.js, Express.js (REST API Architecture)
Database	MongoDB (Mongoose ODM) — <code>awd_shoes_db</code>
Payment Gateway	Razorpay Online Payment Integration
Security & Auth	JSON Web Tokens (JWT), Bcrypt.js Password Encryption, SVG Security CAPTCHA
Target Audience	Footwear retail customers, sports shoe enthusiasts, and store management staff
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Guided By	Minal Desai

2 ENVIRONMENT DESCRIPTION

The environment description outlines the technical setup, hardware requirements, and software stack used to develop and execute the Online Shoes Website.

2.1 Hardware & Software Requirements

2.1.1 Hardware Requirements

The hardware specifications needed to run the client and server components smoothly are listed below:

- **Processor:** Intel Core i3 / AMD Ryzen 3 or higher.
- **Memory (RAM):** Minimum 4 GB RAM (8 GB recommended for running Node server and Vite client concurrently).
- **Hard Disk Storage:** Minimum 500 MB free hard disk space for code files, dependencies (node_modules), and uploaded product images.
- **Display Resolution:** 1366 × 768 pixels or higher for standard desktop and laptop screens.
- **Network / Internet:** Active internet connection required for Google Fonts, brand logos, and Razorpay API communication.

2.1.2 Software Requirements

The software components, tools, and runtimes used in the development environment include:

- **Operating System:** Microsoft Windows 10 / Windows 11 (64-bit).
- **Web Browser:** Google Chrome (v115+), Microsoft Edge, or Mozilla Firefox with JavaScript enabled.
- **Runtime Environment:** Node.js (v18.x, v20.x, or v24.x LTS).
- **Package Manager:** npm (Node Package Manager) for managing project dependencies.
- **Database Engine:** MongoDB Community Server (v6.x or v7.x) running locally on port 27017.
- **Database Management Tool:** MongoDB Compass and MongoDB Shell (mongosh).
- **Code Editor / IDE:** Visual Studio Code / Antigravity IDE.
- **Version Control:** Git & GitHub repository for source code versioning.

2.2 Technology Used

The system is built on modern web development technologies following the MERN (MongoDB, Express.js, React, Node.js) full-stack paradigm:

1. React.js (Frontend User Interface)

React is an open-source component-based JavaScript library used for building interactive single-page user interfaces. In this project, React manages reusable UI components such as shoe product cards, navigation

bars, brand carousels, and modal dialogs. Client-side routing is handled using `react-router-dom` (v6) allowing fast page transitions without page reloads.

2. Vite (Frontend Build Tool)

Vite provides an extremely fast local development server with Hot Module Replacement (HMR) and bundles optimized static production assets using Rollup.

3. Context API (State Management)

React Context API is used for global state management across the application:

- **AuthContext:** Maintains user authentication status, user profile, and JWT token in `localStorage`.
- **CartContext:** Handles cart items, shoe size selection, quantity adjustments, and total price calculation.
- **WishlistContext:** Manages customer-saved favorite shoes for future shopping.

4. Node.js & Express.js (Backend REST API)

Node.js executes JavaScript on the server side using an asynchronous, event-driven engine. Express.js is a minimal and flexible web application framework that provides robust routing, JSON request parsing, and error-handling middleware for all API endpoints (e.g., `/api/auth`, `/api/products`, `/api/orders`, `/api/categories`).

5. MongoDB & Mongoose (Database & ODM)

MongoDB is a document-oriented NoSQL database that stores data in flexible JSON-like documents. Mongoose is used as the Object Data Modeling (ODM) library to enforce strict schemas, relationships, default values, and data validation rules for all entities.

6. Razorpay Payment Gateway

Razorpay provides an online checkout interface supporting UPI, Credit/Debit cards, Net Banking, and digital wallets for completing footwear purchases securely.

3 SYSTEM REQUIREMENT SPECIFICATION

This section presents the structural data requirements, feasibility study, Entity-Relationship model, and UML diagrams representing the Online Shoes Website.

3.1 Data Dictionary

The data dictionary documents the exact collections, field names, data types, constraints, and descriptions used in the MongoDB database (awd_shoes_db).

Table 1: User Collection (users)

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique MongoDB identifier for the user account
name	String	Required	Full name of the customer
email	String	Required, Unique	Email address used for authentication and communications
password	String	Required	Bcrypt-hashed encrypted account password
phone	String	Optional	Contact mobile number of the customer
address	String	Optional	Residential/shipping street address
role	String	Default: 'user'	Account privilege level ('user' or 'admin')
createdAt	Date	Auto-generated	Timestamp of account registration

Table 2: Product Collection (products)

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique MongoDB identifier for the shoe item
name	String	Required	Model name of the shoe (e.g. Nike Air Max)
brand	String	Required	Brand name (Nike, Puma, Adidas, Converse, etc.)
category	ObjectId	Required, FK	Reference to Category collection _id
price	Number	Required	Retail price in Indian Rupees (INR)
stock	Number	Required, Default: 10	Available inventory stock quantity
sizes	Array of Numbers	Default: [6,7,8,9,10,11]	Available shoe sizes (UK sizing)
description	String	Required	Shoe specifications, material, and cushioning details

Field Name	Data Type	Constraint	Description
image	String	Required	File path / URL of product image in uploads folder
isFeatured	Boolean	Default: false	Flag to display shoe in Featured section on Home page

Table 3: Category Collection (categories)

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique MongoDB identifier for category
name	String	Required, Unique	Category title (Running, Casual, Sports, Formal, Boots)
description	String	Optional	Short summary explaining footwear purpose

Table 4: Brand Collection (brands)

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique MongoDB identifier for brand
name	String	Required, Unique	Brand name (Nike, Adidas, Puma, Skechers, etc.)
slug	String	Optional	URL-friendly brand slug
logo	String	Optional	Relative path to brand vector logo icon
description	String	Optional	Short brand profile summary

Table 5: Order Collection (orders)

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique order tracking identifier
user	ObjectId	Required, FK	Reference to customer in User collection
items	Array of Objects	Required	Ordered shoes: product ID, name, size, price, qty
shippingAddress	Object	Required	Full name, phone, street address, city, postal code
paymentMethod	String	Default: 'Razorpay'	Selected payment gateway
paymentStatus	String	Default: 'Pending'	Payment result ('Pending' or 'Paid')
razorpayPaymentId	String	Optional	Transaction ID issued by Razorpay upon success

Field Name	Data Type	Constraint	Description
totalAmount	Number	Required	Final invoice amount paid in INR
status	String	Default: 'Pending'	Fulfillment status: Pending, Processing, Shipped, Delivered

Table 6: Review Collection (reviews)

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique review identifier
product	ObjectId	Required, FK	Reference to reviewed shoe product
user	ObjectId	Required, FK	Reference to reviewer customer account
userName	String	Required	Display name of customer
rating	Number	Required (1 to 5)	Customer star rating for comfort and quality
comment	String	Required	Written feedback and comments

Table 7: ContactMessage Collection (contactmessages)

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique message identifier
name	String	Required	Name of customer sending inquiry
email	String	Required	Reply email address
subject	String	Required	Inquiry subject topic
message	String	Required	Full inquiry text message

Table 8: Admin Collection (admins)

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique admin identifier
username	String	Required	Admin portal login username
email	String	Required, Unique	Admin contact and notification email
password	String	Required	Encrypted password for administrator credentials


```
MongoDB Shell (mongosh v2.1.0) – Database: awd_shoes_db – Host: localhost:27017

test> use awd_shoes_db
switched to db awd_shoes_db

awd_shoes_db> show collections
admins
brands
categories
contactmessages
orders
products
reviews
users

awd_shoes_db> db.categories.find().pretty()
[
  {
    "_id": "6a96f5e3bc23d5a46743ecd0",
    "name": "Casual Sneakers",
    "description": "Stylish and comfortable sneakers for everyday casual wear.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd1",
    "name": "Formal Shoes",
    "description": "Classic handcrafted leather formal shoes for business & events.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd2",
    "name": "Athletic Shoes",
    "description": "High-performance athletic shoes for running and sports.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd3",
    "name": "Boat Shoes",
    "description": "Casual boat shoes with non-slip soles for nautical adventures.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd4",
    "name": "Loafers",
    "description": "Elegant loafers for formal occasions and business meetings.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd5",
    "name": "Sandals",
    "description": "Comfortable sandals for warm weather and casual outings.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd6",
    "name": "Slippers",
    "description": "Cozy slippers for indoor comfort and relaxation.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd7",
    "name": "Socks",
    "description": "High-quality socks for comfort and durability.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd8",
    "name": "Shoe Care Products",
    "description": "Essential shoe care products for maintaining your footwear.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  },
  {
    "_id": "6a96f5e3bc23d5a46743ecd9",
    "name": "Shoe Accessories",
    "description": "Stylish shoe accessories to complement your footwear.",
    "__v": 0,
    "createdAt": "2026-09-01T15:57:23.391Z",
    "updatedAt": "2026-09-01T15:57:23.391Z"
  }
]
```

Figure 3.1: Actual MongoDB Database (awd_shoes_db) Shell & Collection View

3.2 Feasibility Analysis

A feasibility study is conducted to evaluate whether the proposed system can be successfully developed, implemented, and utilized within practical college project limits.

3.2.1 Economic Feasibility

Economic feasibility examines the financial investment needed to develop and operate the system:

- **Development Tools Cost:** The entire technology stack (Node.js, Express, React, MongoDB, VS Code) is 100% free and open-source. No commercial licenses or proprietary tools were purchased.
- **Hardware Cost:** Existing standard college lab computers and personal laptops were used for development without requiring specialized server hardware.
- **Operational & Maintenance Cost:** Routine database backups and code updates can be performed with minimal computing resources. Cloud hosting platforms (such as Render, Vercel, and MongoDB Atlas) provide free tier options suitable for student projects.
- **Conclusion:** The project has virtually zero monetary cost and is economically completely feasible.

3.2.2 Technical Feasibility

Technical feasibility determines whether the required technical skills and technologies are available:

- **MERN Stack Compatibility:** Node.js, Express, MongoDB, and React work seamlessly together using JSON as a universal data exchange format.
- **Developer Skill Set:** The team has acquired practical knowledge in HTML5, CSS3, JavaScript, React components, RESTful routing, and MongoDB querying through the BCA Semester 5 curriculum.
- **Reliability:** MongoDB provides reliable document storage with atomic operations, and React provides responsive client-side state handling.
- **Conclusion:** The project is technically sound and fully feasible.

3.2.3 Behavioural Feasibility

Behavioural feasibility assesses user acceptance and ease of learning:

- **Customer Acceptance:** The website follows standard e-commerce layout conventions (search bar, cart badge, size selectors, clear checkout). Users require no prior training to shop.
- **Admin Usability:** The admin dashboard has straightforward forms for adding shoes and updating orders without requiring technical database commands.
- **Conclusion:** The system is user-friendly and behaviourally feasible.

3.3 E-R Diagram

The Entity-Relationship (E-R) diagram models the conceptual database architecture of the Online Shoes Website. It defines the major entities (User, Product, Category, Brand, Order, Review, ContactMessage, Admin) and the cardinality of relationships between them.

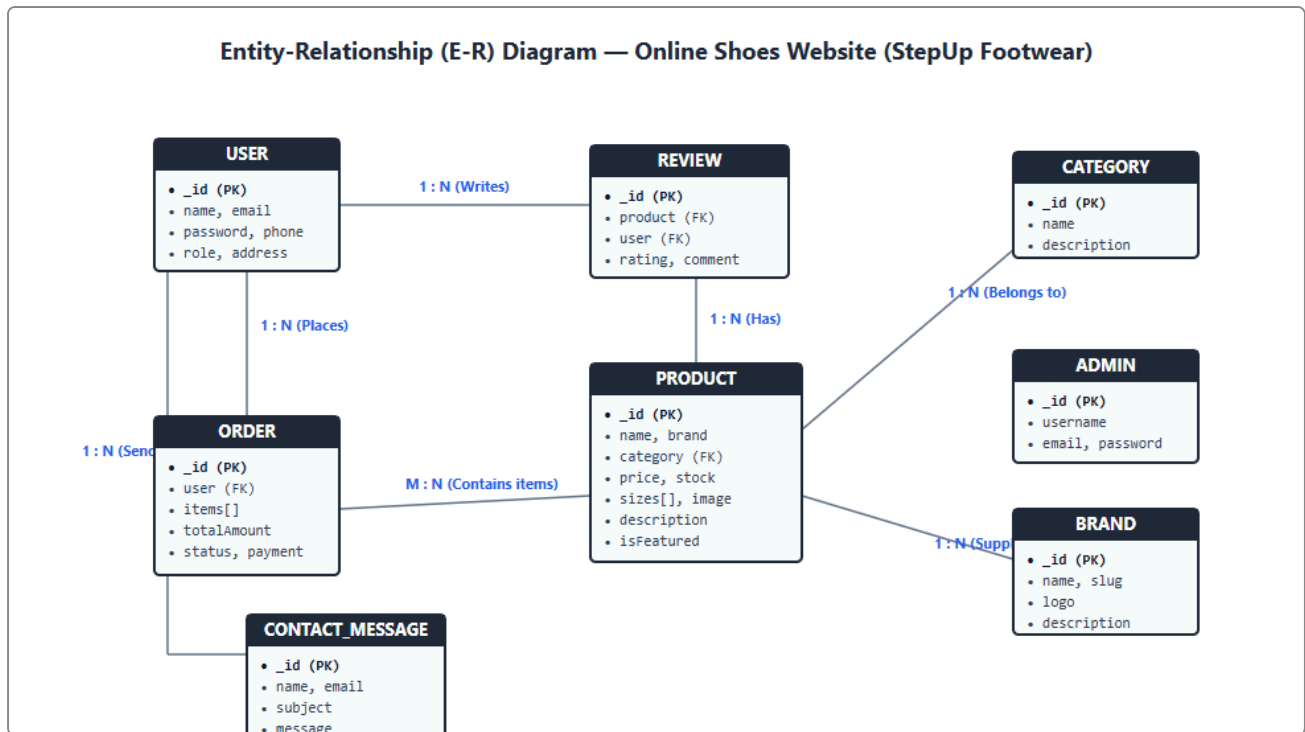


Figure 3.2: Entity-Relationship (E-R) Diagram of Online Shoes Website

Relationship Explanations:

- **User — Order (1 : N)**: A single registered customer can place multiple shoe orders over time. Each order belongs strictly to one customer.
- **User — Review (1 : N)**: A customer can write multiple product reviews and ratings. Each review is linked to one user.
- **Category — Product (1 : N)**: Each shoe category (e.g. Running, Casual, Formal) groups multiple shoe models. Each shoe product belongs to one primary category.
- **Brand — Product (1 : N)**: A footwear brand (e.g. Nike, Puma, Adidas) manufactures multiple shoe products in the catalog.
- **Product — Review (1 : N)**: Each shoe item can receive multiple customer reviews and star ratings.
- **Order — Product (M : N via Order Items)**: An order can contain multiple pairs of shoes of various sizes, and a shoe product can be part of many customer orders.
- **User — ContactMessage (1 : N)**: A customer can submit multiple support inquiries or messages.

3.4 UML Diagrams

Unified Modeling Language (UML) diagrams provide standard visual representations of system actors, use cases, classes, and workflows.

3.4.1 Use Case Diagram

The Use Case diagram identifies the primary actors (Customer and Store Administrator) and their direct interactions with the system modules.

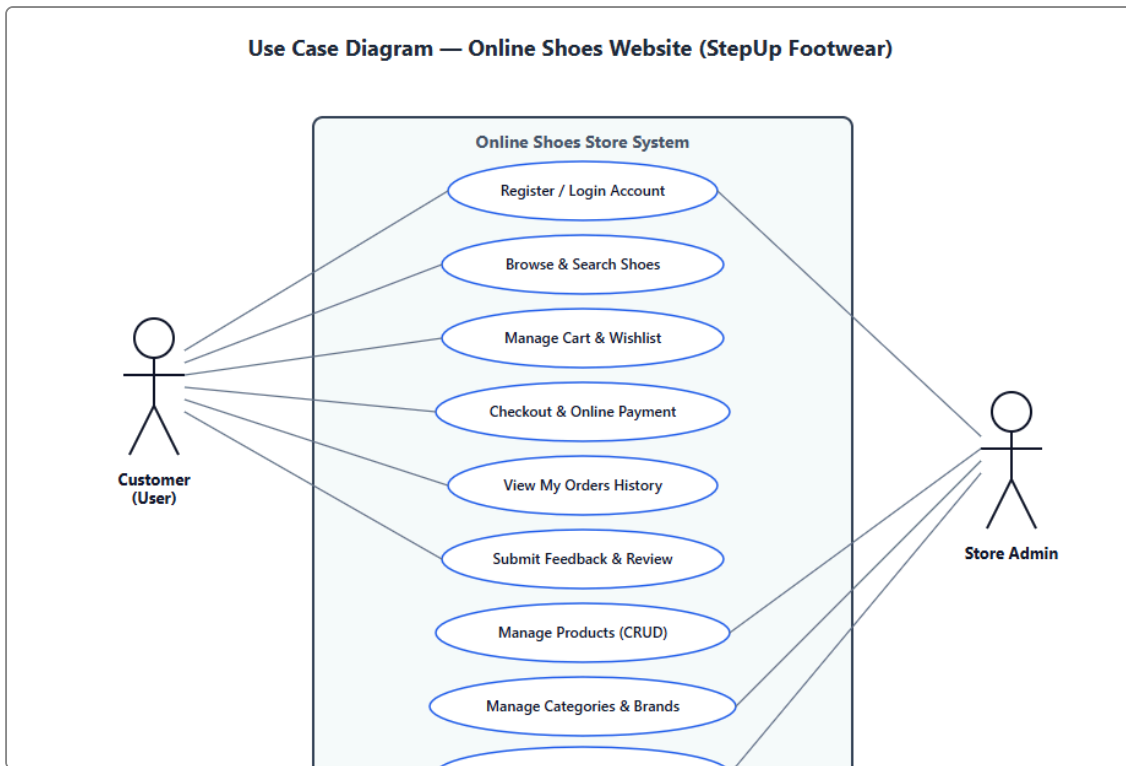


Figure 3.3: Use Case Diagram for Online Shoes Website

3.4.2 Class Diagram

The Class diagram models the object-oriented structure of the application, detailing the class attributes and member methods implemented across the models and controllers.

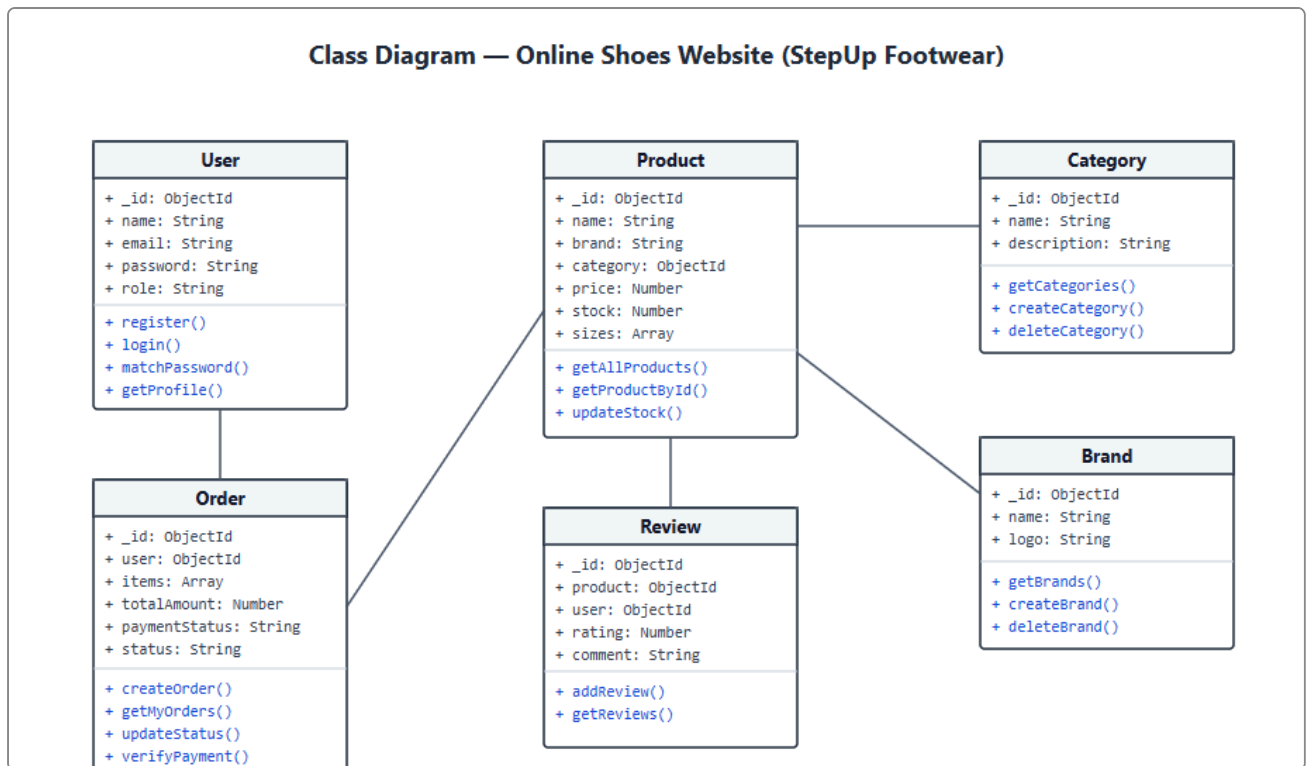


Figure 3.4: Class Diagram for Online Shoes Website

4 SOFTWARE PROJECT PLANNING

Software project planning defines the functional boundaries, core goals, and practical benefits expected from the Online Shoes Website.

4.1 Scope

The scope of the Online Shoes Website encompasses full e-commerce operations tailored for footwear retail:

- **Customer Account Management:** Account registration, secure login with security CAPTCHA, profile management, and order history view.
- **Catalog Exploration:** Real-time shoe search by model name, category-based filtering, brand-based filtering, and price range sorting.
- **Product Details & Sizing:** Detailed shoe descriptions, multi-angle images, available stock counts, and UK shoe size selection (Sizes 6 to 11).
- **Cart & Wishlist:** Real-time cart calculation with quantity increment/decrement, persistent wishlist storage, and item removal.
- **Checkout & Payment:** Delivery address collection and online payment integration via Razorpay.
- **Customer Feedback:** Star ratings (1 to 5 stars) and comments on shoe performance and fit.
- **Independent Administrator Portal:** Isolated admin dashboard displaying live order metrics, product management (add/edit/delete shoes), category management, brand listings, order fulfillment status updates, and user inquiry reviews.

4.2 Objectives

The primary objectives of developing the Online Shoes Website are:

- To replace traditional paper-based shoe sales registers with an automated database.
- To provide customers with 24/7 access to browse and purchase shoes from anywhere.
- To provide transparent shoe sizing information and accurate stock availability.
- To streamline order tracking so customers can check order progress (Pending, Processing, Shipped, Delivered).
- To secure customer data using industry-standard password hashing (Bcrypt) and stateless token authentication (JWT).
- To prevent bot attacks on login and registration through mathematical/alphanumeric CAPTCHA verification.
- To give store managers an easy-to-use admin console for inventory and sales oversight.

4.3 Expected Advantages

The practical advantages of the implemented system include:

- **Time Efficiency:** Customers save travel time and can place shoe orders in less than two minutes.
- **Reduced Human Error:** Automated bill calculation eliminates arithmetic errors in subtotals and discounts.
- **Real-Time Inventory Control:** Stock counts update immediately upon order placement, preventing overselling.
- **Better Customer Engagement:** Real customer reviews and star ratings build trust and help shoppers pick the right shoes.
- **Centralized Data Storage:** All orders, customer details, and product catalogs are safely stored in MongoDB with simple backup capabilities.

5 SYSTEM DESIGN

System design covers the input forms through which users supply information to the system and the corresponding output interfaces displayed to users and administrators.

5.1 Input Design

Input design ensures that data entered into the system is accurate, validated, and user-friendly. Real screenshots of the implemented input forms are provided below:

1. Customer Sign In Form

The customer login form requires an email address, password, and security CAPTCHA verification. Form inputs are pre-filled with demo credentials for easy faculty evaluation.

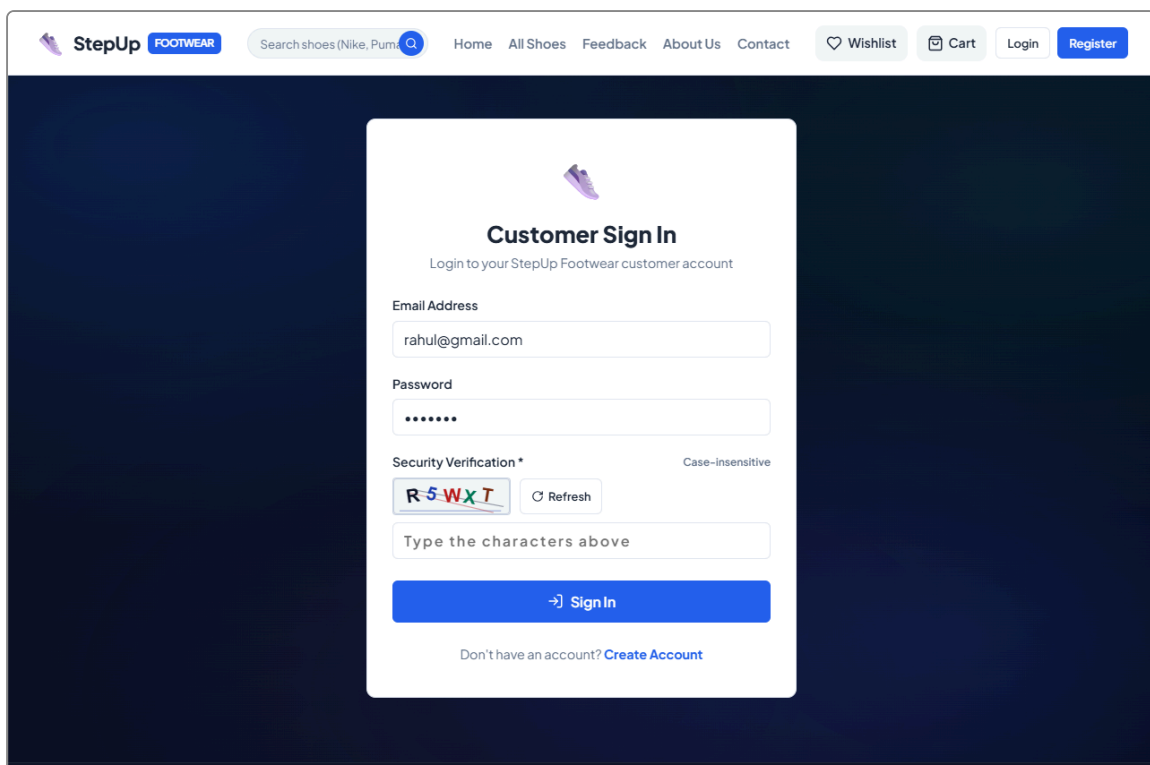
The image shows a web browser window displaying the 'StepUp FOOTWEAR' website. The main content area features a 'Customer Sign In' form. The form has a title 'Customer Sign In' with a shoe icon above it, and a subtitle 'Login to your StepUp Footwear customer account'. It contains three input fields: 'Email Address' (pre-filled with 'rahul@gmail.com'), 'Password' (pre-filled with '*****'), and 'Security Verification *' (a CAPTCHA showing the characters 'R-5-W-X-T' with a 'Refresh' button). Below the CAPTCHA is a text input field with the placeholder 'Type the characters above'. A blue 'Sign In' button is at the bottom of the form, and a link 'Don't have an account? Create Account' is below it. The website's header includes a search bar, navigation links (Home, All Shoes, Feedback, About Us, Contact), and user links (Wishlist, Cart, Login, Register).

Figure 5.1: Customer Sign In Form with CAPTCHA

2. Customer Registration Form

Allows new customers to create an account by providing full name, email, contact phone, delivery address, password, and CAPTCHA.

StepUp FOOTWEAR Search shoes (Nike, Puma) Home All Shoes Feedback About Us Contact Wishlist Cart Login Register

Create an Account

Join StepUp Footwear to shop the best shoe collection

Full Name *
e.g. Ramesh Patel

Email Address * **Phone Number**
name@example.com 9876543210

Password (min 6 chars) * **Confirm Password ***

Delivery Address
Your street address, city, pincode

Security Verification * Case-insensitive
M J 4 S T Refresh
Type the characters above

Figure 5.2: Customer Registration Form

3. Checkout & Delivery Address Form

Collects delivery recipient details (full name, phone, street address, city, and PIN code) and payment method selection before opening Razorpay checkout.

StepUp FOOTWEAR Search shoes (Nike, Puma) Home All Shoes Feedback About Us Contact Wishlist Cart Login Register

Customer Sign In

Login to your StepUp Footwear customer account

Email Address
rahul@gmail.com

Password

Security Verification * Case-insensitive
P A E R 4 Refresh
Type the characters above

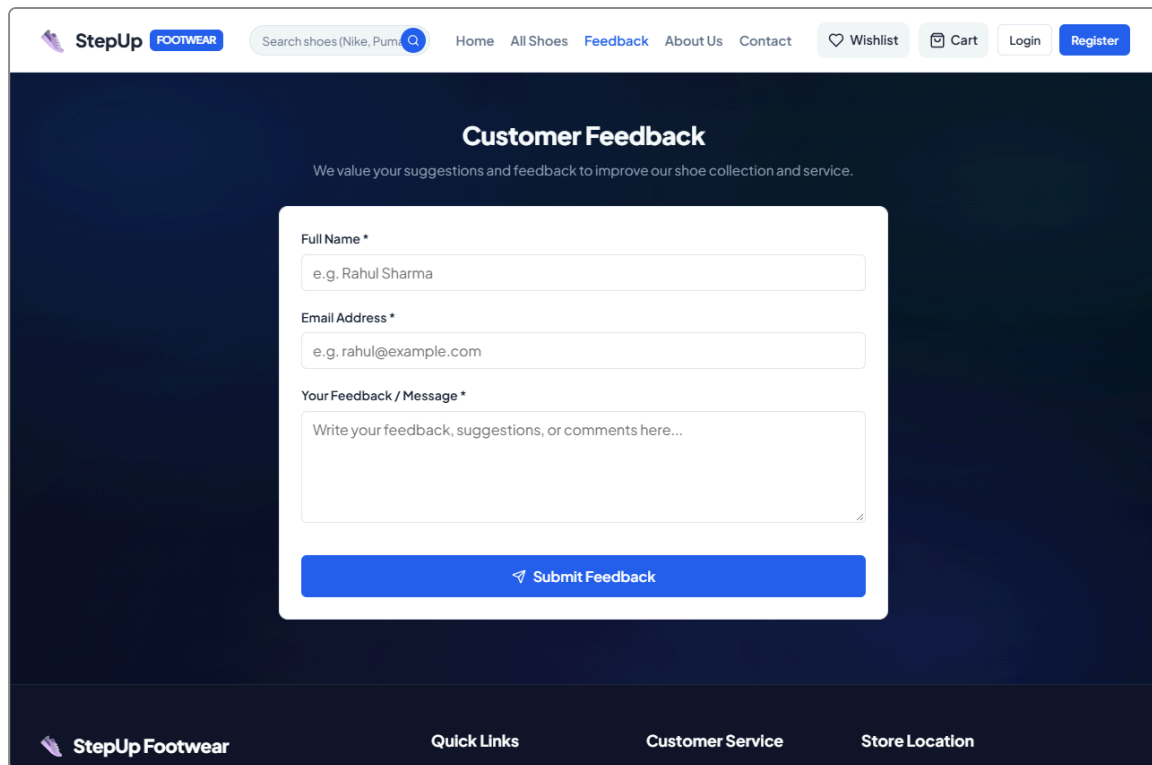
Sign In

Don't have an account? [Create Account](#)

Figure 5.3: Checkout Delivery Address Form

4. Customer Feedback & Review Form

Enables customers to select a shoe product, submit a 1-to-5 star rating, and provide qualitative feedback.

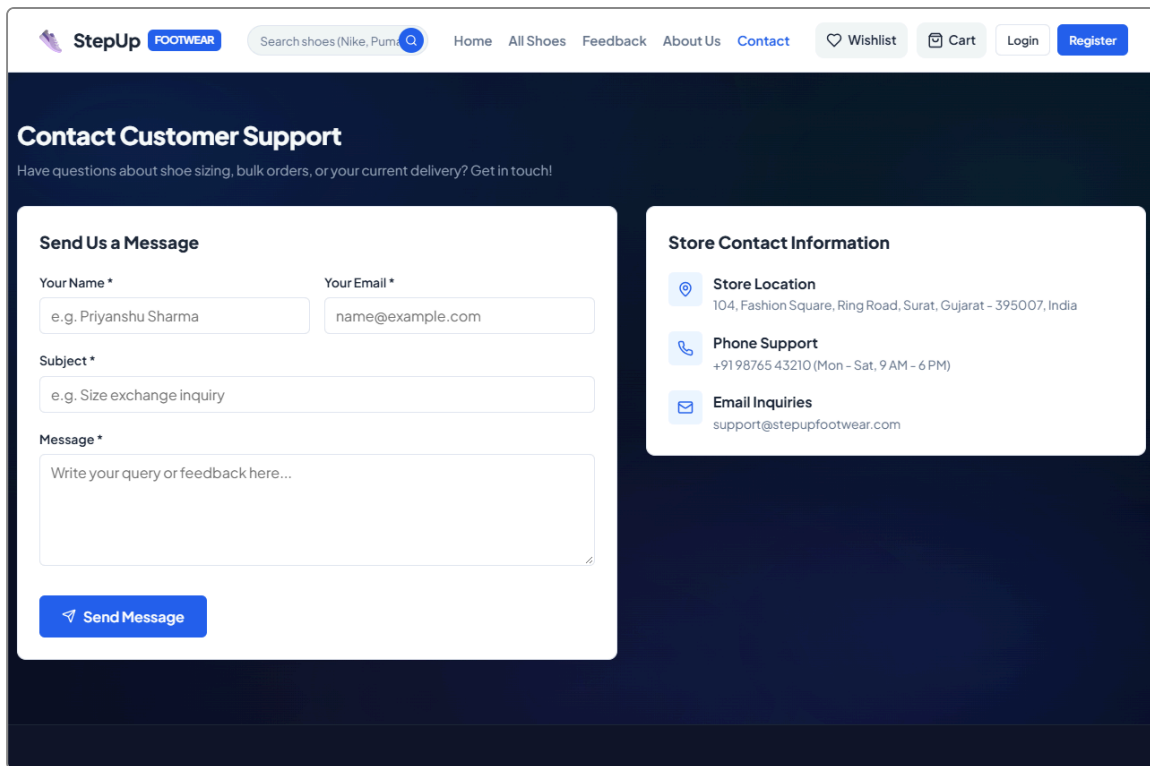


The screenshot displays the 'Customer Feedback' form on the StepUp Footwear website. The form is centered on a dark blue background. It includes a header with the StepUp logo, a search bar, and navigation links. The form itself has three input fields: 'Full Name *' with a placeholder 'e.g. Rahul Sharma', 'Email Address *' with a placeholder 'e.g. rahul@example.com', and 'Your Feedback / Message *' with a placeholder 'Write your feedback, suggestions, or comments here...'. A blue 'Submit Feedback' button is at the bottom of the form. The footer contains the StepUp Footwear logo and three quick links: 'Quick Links', 'Customer Service', and 'Store Location'.

Figure 5.4: Customer Feedback & Review Form

5. Contact Support Form

Provides a direct channel for customer inquiries regarding sizing, bulk orders, or shipment delays.

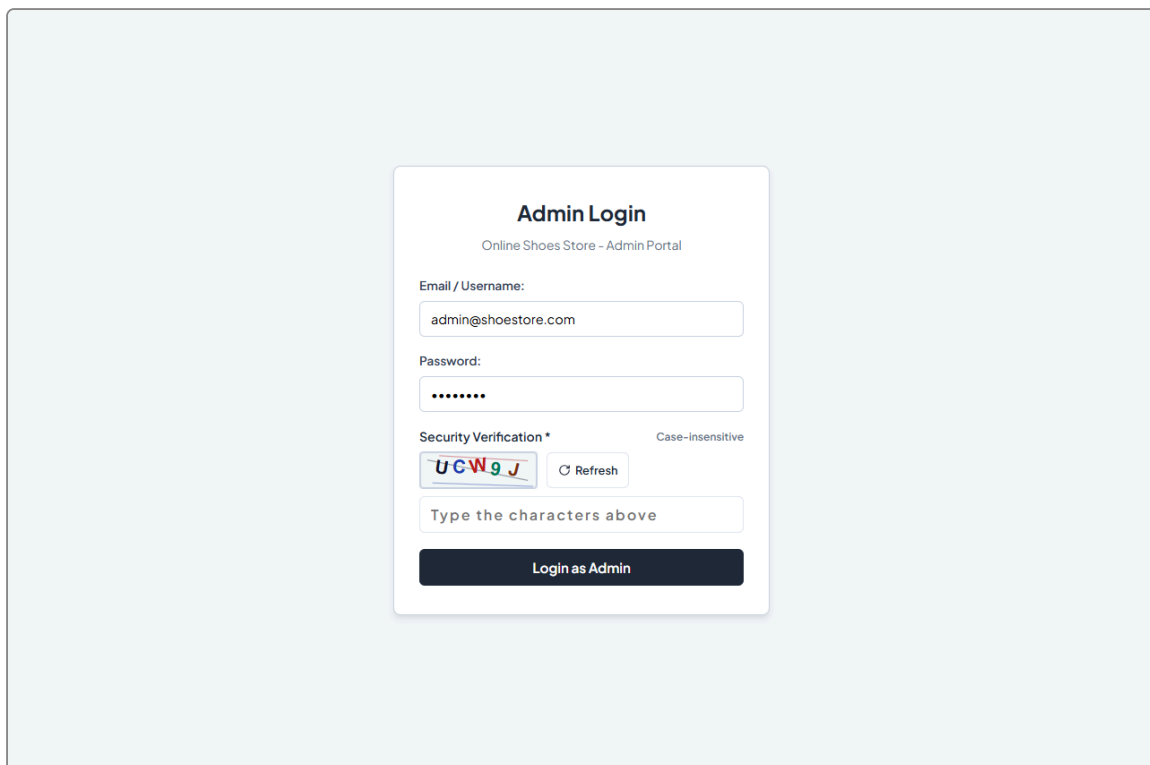


The image shows a web interface for 'StepUp FOOTWEAR'. At the top is a navigation bar with a search bar (containing 'Search shoes (Nike, Puma)'), links for Home, All Shoes, Feedback, About Us, and Contact, and buttons for Wishlist, Cart, Login, and Register. The main section is titled 'Contact Customer Support' with a subtitle 'Have questions about shoe sizing, bulk orders, or your current delivery? Get in touch!'. It contains two white boxes on a dark blue background. The left box, 'Send Us a Message', has fields for 'Your Name *' (filled with 'e.g. Priyanshu Sharma'), 'Your Email *' (filled with 'name@example.com'), 'Subject *' (filled with 'e.g. Size exchange inquiry'), and a 'Message *' text area (filled with 'Write your query or feedback here...'). A blue 'Send Message' button is at the bottom. The right box, 'Store Contact Information', lists: 'Store Location' (104, Fashion Square, Ring Road, Surat, Gujarat - 395007, India), 'Phone Support' (+91 98765 43210 (Mon - Sat, 9 AM - 6 PM)), and 'Email Inquiries' (support@stepupfootwear.com).

Figure 5.5: Contact Support Inquiry Form

6. Administrator Login Form

Dedicated admin login screen isolated from the client website, featuring pre-filled admin credentials and CAPTCHA validation.



The image shows a dedicated 'Admin Login' form centered on a light blue background. The form is titled 'Admin Login' with the subtitle 'Online Shoes Store - Admin Portal'. It contains three input fields: 'Email / Username:' (pre-filled with 'admin@shoestore.com'), 'Password:' (masked with '*****'), and 'Security Verification *'. The security verification includes a CAPTCHA image showing the characters 'UCW9J' and a 'Refresh' button. Below the CAPTCHA is a text input field labeled 'Type the characters above'. At the bottom is a dark blue 'Login as Admin' button.

Figure 5.6: Dedicated Administrator Login Form

7. Admin Category Management Form

Allows administrators to create new footwear categories and edit existing category names and descriptions.

#	Category Name	Description	Created At	Actions
1	Casual Sneakers	Stylish and comfortable sneakers for everyday casual wear.	1/9/2026	
2	Formal Shoes	Classic handcrafted leather formal shoes for business & events.	1/9/2026	
3	Loafers & Slip-ons	Comfortable slip-on shoes and elegant loafers.	1/9/2026	
4	Outdoor & Boots	Sturdy all-weather trekking boots and high-grip outdoor shoes.	1/9/2026	
5	Running Shoes	Lightweight and cushioned running shoes for daily performance.	1/9/2026	
6	Sports & Training	Durable and supportive athletic footwear for fitness & sports.	1/9/2026	

Figure 5.7: Admin Category Creation Form

8. Admin Shoe Product Form

Allows the store manager to add new shoe models with model name, brand, category, price, stock count, description, and image upload.

Admin Panel

Dashboard
Categories
Brands
Products
Orders
Users
Inquiries
Reviews
Back to Store
Logout

Product Management (CRUD)

Add, edit, upload shoe images, and update footwear catalog details

+ Add New Shoe

All Shoes in Store (8)

Image	Shoe Name	Brand	Category	Price	Stock	Featured	Actions
	Nike Zoom Speed Pro Running Shoe Sizes: 7, 8, 9, 10, 11	Nike	Running Shoes	₹3,499	11 pairs	Featured	 
	Classic Urban White Leather Sneakers Sizes: 6, 7, 8, 9, 10	Adidas	Casual Sneakers	₹2,499	19 pairs	Featured	 
	Royal Heritage Brown Oxford Shoes Sizes: 7, 8, 9, 10	Clarks	Formal Shoes	₹3,999	11 pairs	Featured	 
	AeroGlide Ultra Blue Sports Shoes Sizes: 7, 8, 9, 10, 11	Puma	Running Shoes	₹2,899	17 pairs	Featured	 
	Milano Suede Tan Slip-on Loafers Sizes: 7, 8, 9, 10	Woodland	Loafers & Slip-ons	₹2,799	14 pairs	No	 
	Apex Trail Waterproof Hiking Boots Sizes: 8, 9, 10, 11	Woodland	Outdoor & Boots	₹4,499	10 pairs	No	 
	Streetwear Canvas Low-Top Sneakers	Converse	Casual	₹1,899	24	Featured	 

Figure 5.8: Admin Shoe Product Management Form

5.2 Output Design

Output design focuses on presenting information clearly and effectively to both customers and store administrators. Real screenshots of the output screens are presented below:

1. Home Page & Brand Marquee

Displays the store banner, featured footwear collections, brand logos (Nike, Puma, Adidas, Converse, Skechers), and special promotions.

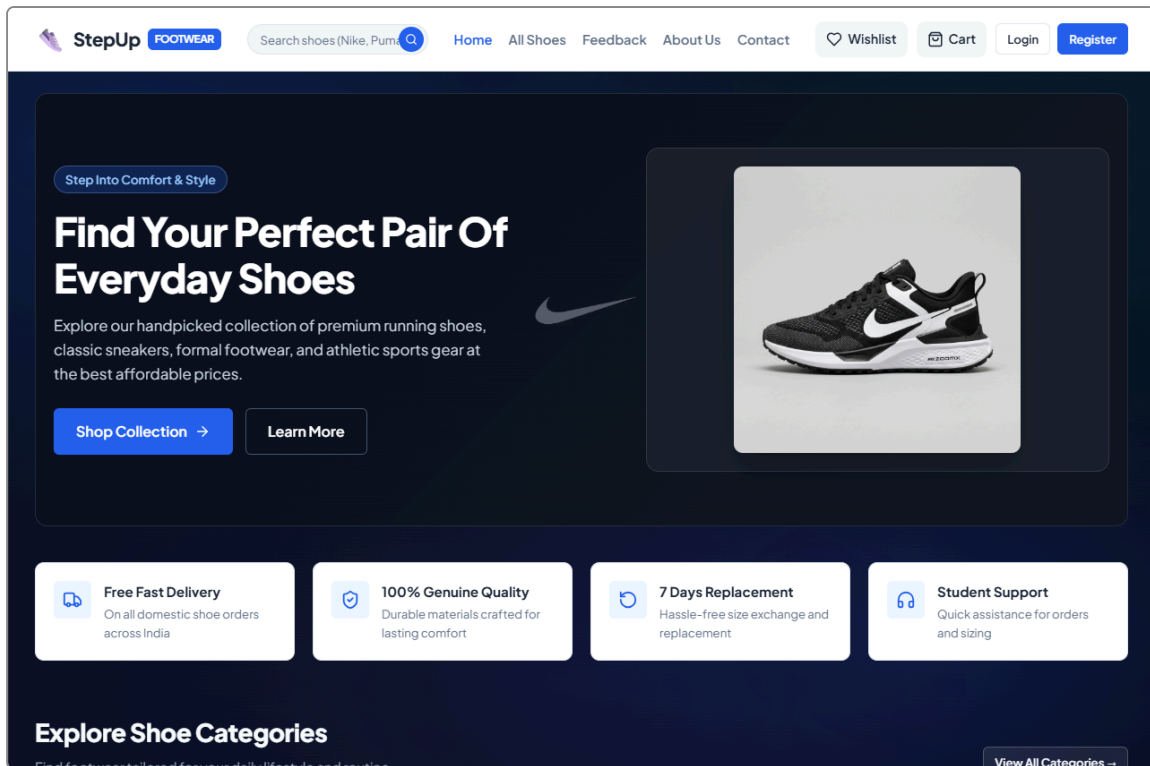


Figure 5.9: Home Page with Brand Showcase

2. All Shoes Catalog with Filters

Displays the full shoe catalog in a responsive grid with category filter tags, brand filter buttons, and search results.

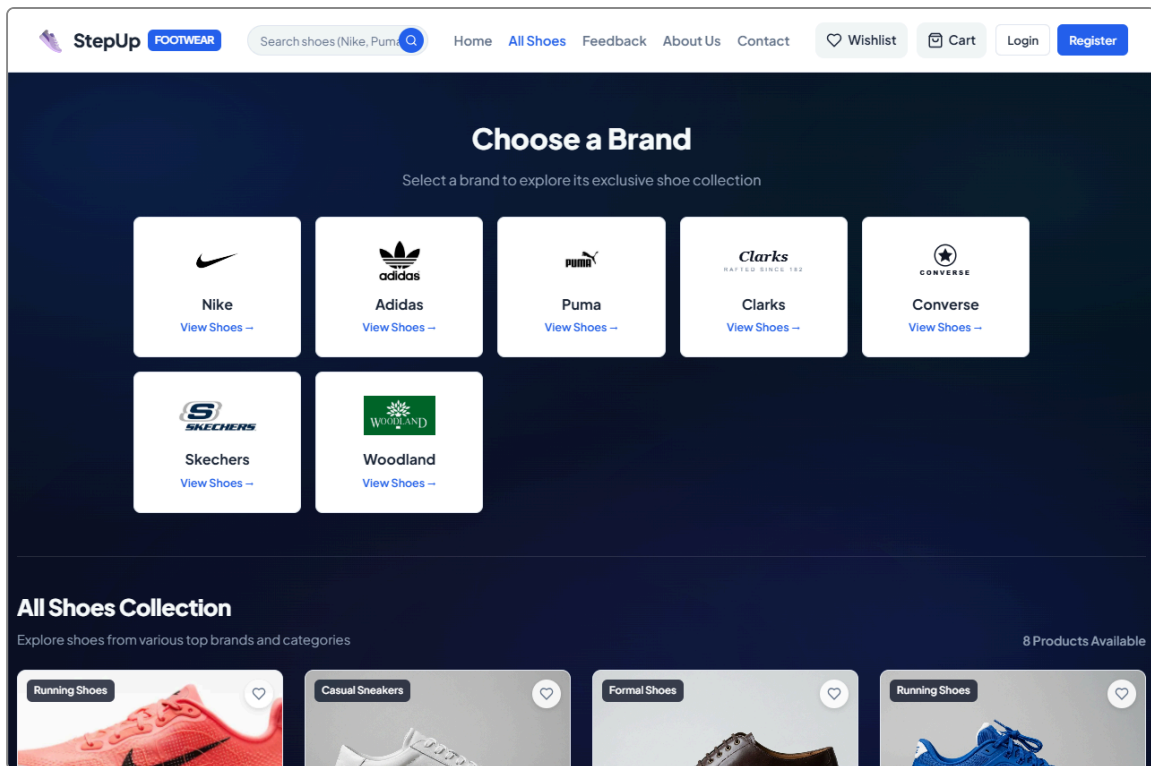


Figure 5.10: All Shoes Catalog with Filtering

3. Product Details Screen

Shows high-resolution shoe imagery, brand badge, retail price, stock status, available UK sizes (6 to 11), and "Add to Cart" / "Add to Wishlist" buttons.

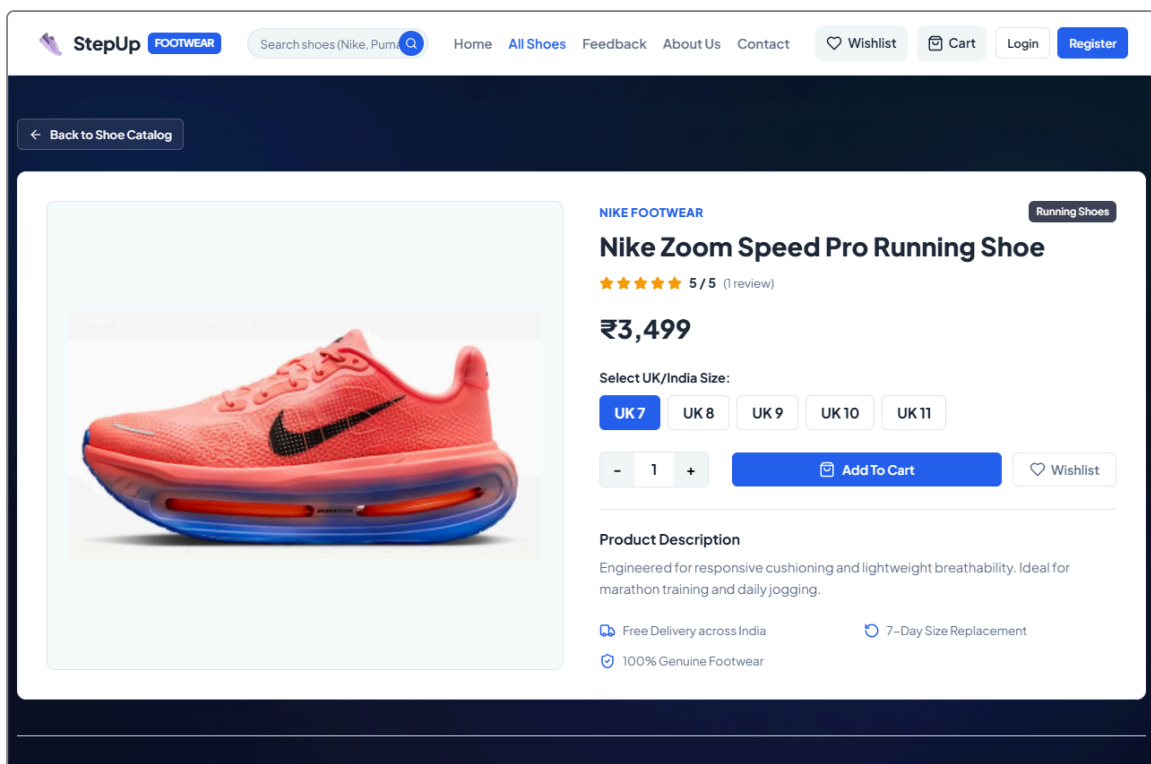


Figure 5.11: Shoe Product Details Screen

4. Shopping Cart Screen

Lists selected shoes, chosen sizes, item quantity adjusters, individual item subtotals, free delivery indicator, and "Proceed to Checkout" button.

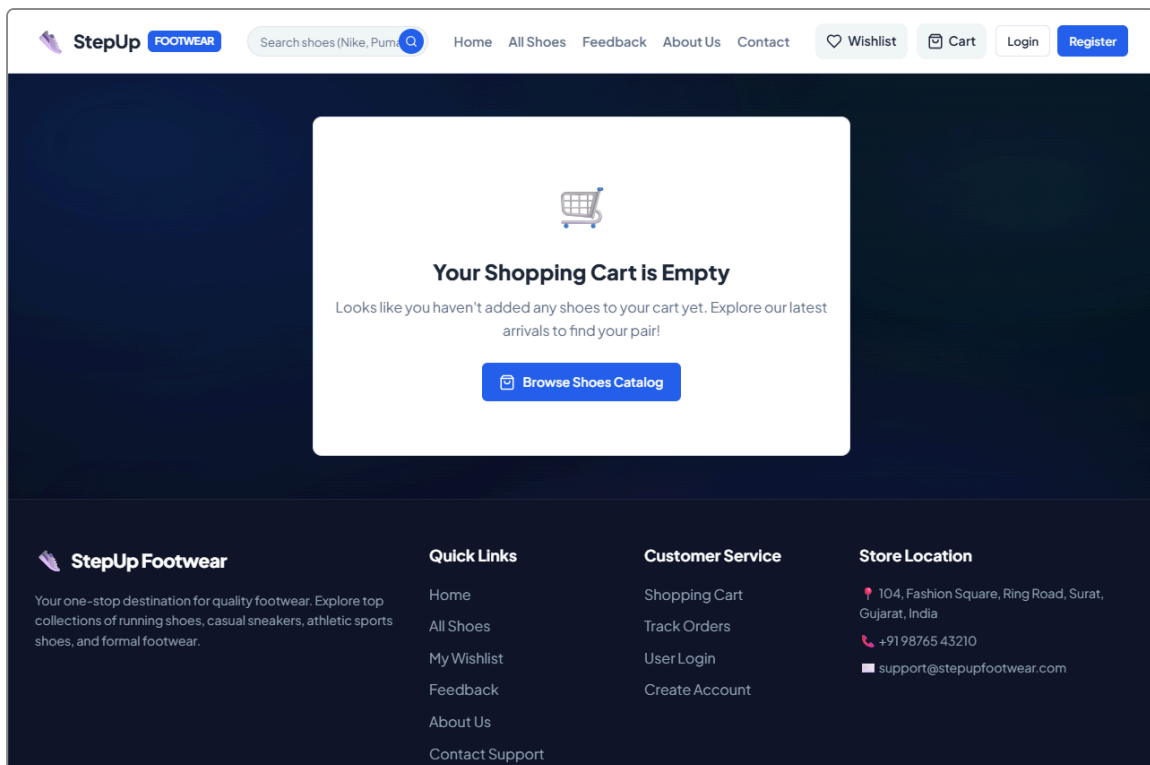


Figure 5.12: Shopping Cart Overview Screen

5. Admin Dashboard & Analytics

Provides store managers with real-time statistics on total products, active categories, total orders placed, and recent customer orders with serial indexes (#1, #2, #3).

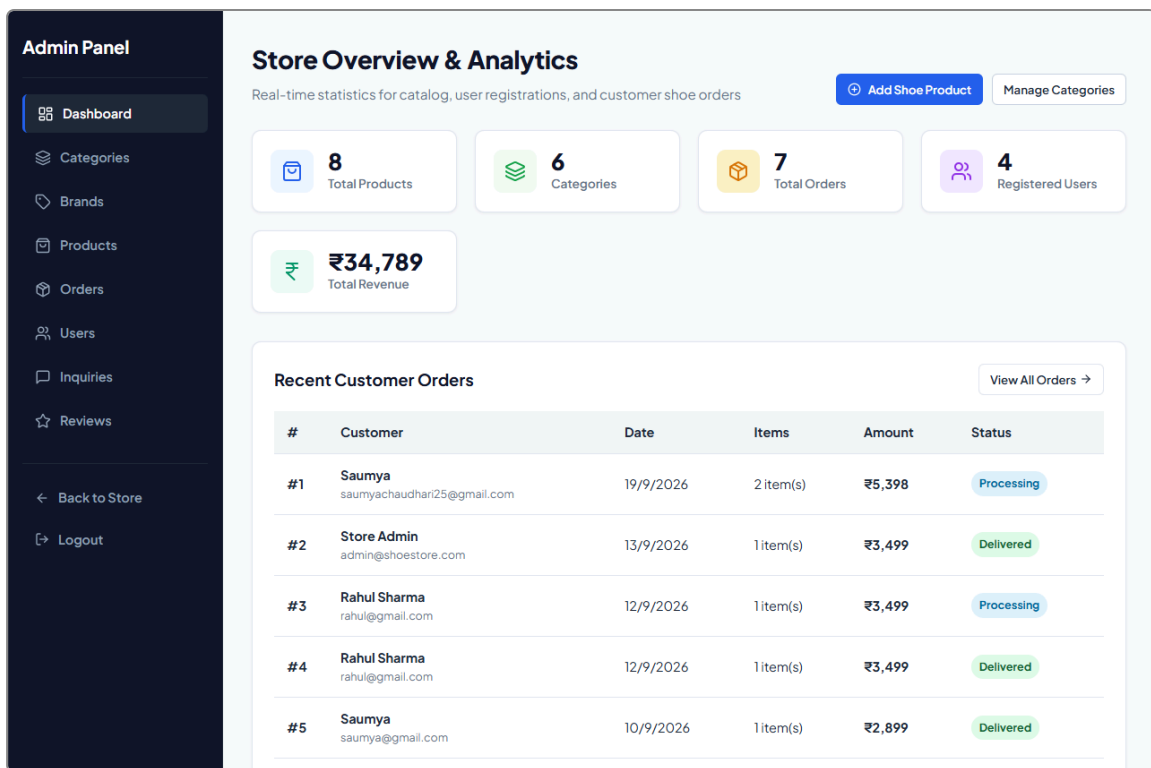


Figure 5.13: Admin Dashboard with Live Statistics

6. Admin Orders Management Screen

Displays all placed orders with customer details, order items, invoice amounts, payment status, and a status dropdown for updating order fulfillment.

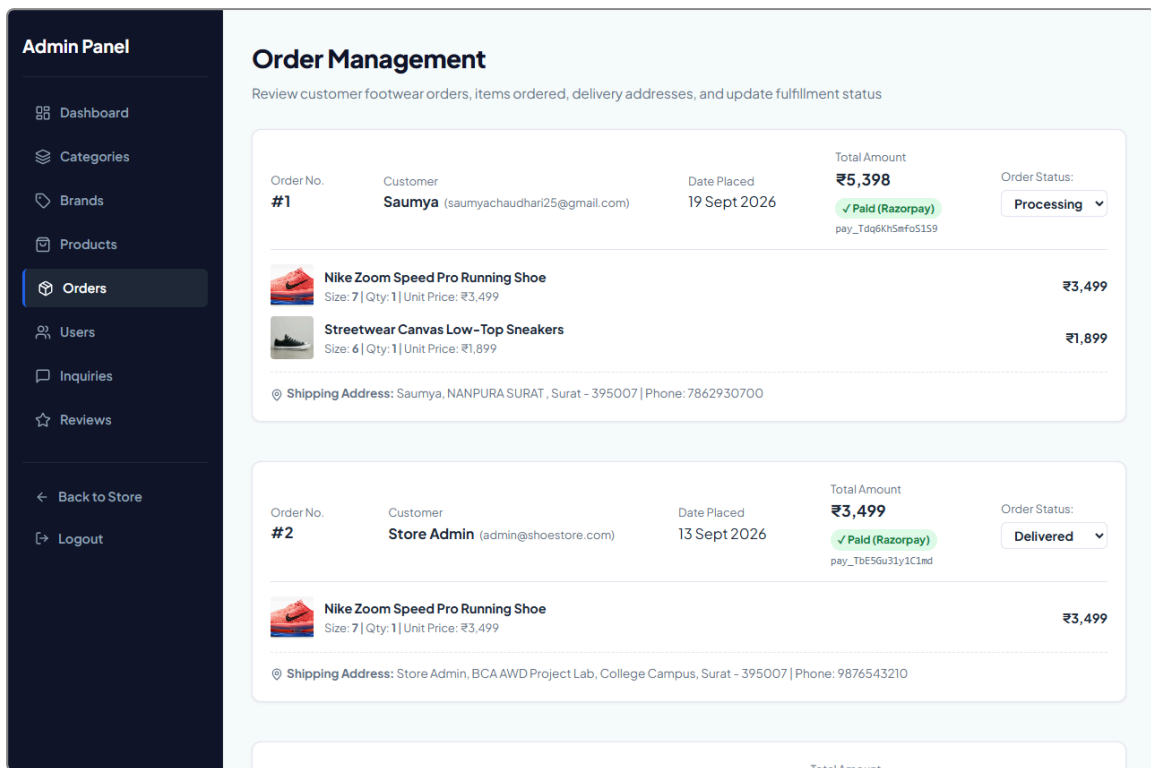


Figure 5.14: Admin Orders Fulfillment Screen

7. Admin Brands Management Screen

Shows all registered footwear brands, their logo representations, and associated product counts.

Admin Panel

Dashboard

Categories

Brands

Products

Orders

Users

Inquiries

Reviews

← Back to Store

→ Logout

Brand Management (CRUD)

Create, update, and manage official shoe brands for the store and catalog

Add New Shoe Brand

Brand Name *

e.g. Reebok, Under Armour, Asics

Brand Description / Tagline

e.g. High performance athletic footwear and running snee

Brand Logo (SVG, PNG, JPG)

Choose File

No file chosen

+ Add Brand

Existing Brands (7)

#	Logo	Brand Name	Slug	Description	Actions
1		Nike	nike	Just Do It — World leading athletic, running & sports footwear.	 
2		Adidas	adidas	Impossible Is Nothing — Iconic lifestyle sneakers and high-performance shoes.	 
3		Puma	puma	Forever Faster — Cutting-edge dynamic trainers and athletic sports shoes.	 
4		Clarks	clarks	Heritage Craftsmanship — Premium handcrafted leather formal and oxford shoes.	 
5		Converse	converse	Timeless Streetwear — Authentic canvas low-top & skate sneakers.	 
6		Skechers	skechers	Comfort Revolution — Ultra-light cushioned walking and cross-training shoes.	 
7		Woodland	woodland	Explore the Outdoors — Heavy-duty trekking boots and	 

Figure 5.15: Admin Brand Management Screen

8. Admin Customer Inquiries Screen

Lists all contact messages received from website visitors with sender name, email, subject, and message content.

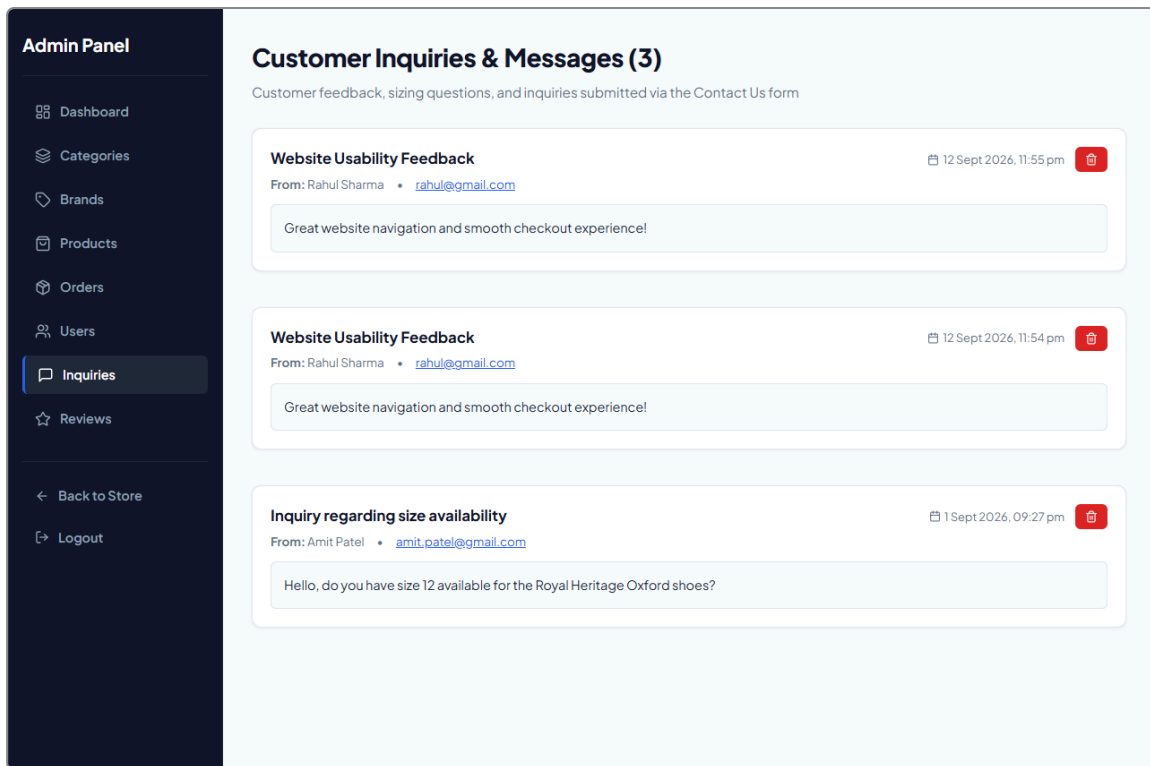


Figure 5.16: Admin Customer Inquiries Screen

9. Admin Shoe Reviews Screen

Displays customer reviews and star ratings with moderation controls for the store administrator.

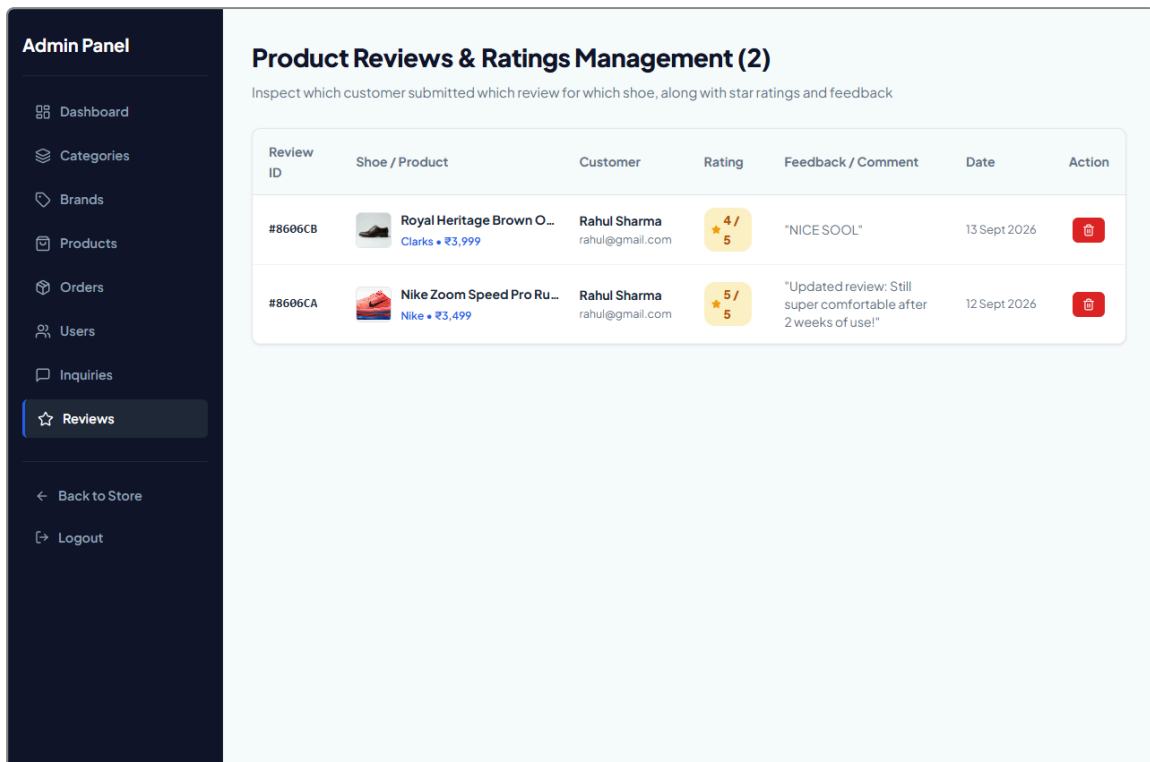


Figure 5.17: Admin Shoe Reviews Management Screen

6 SOFTWARE TESTING

6.1 Introduction

Software testing is a systematic process of executing application components with the intent of finding defects, verifying input validation rules, and confirming that all implemented features operate according to the project requirements. Testing ensures that the Online Shoes Website is reliable, secure, and user-friendly prior to deployment.

6.2 Objectives of Testing

The core objectives of testing this web application include:

- To confirm that user registration, login, and CAPTCHA validation work correctly.
- To verify that shoe search, category filtering, and brand navigation yield accurate results.
- To check that shopping cart calculations (subtotal, quantity adjustments, and total payable) are accurate.
- To test that Razorpay payment checkout triggers properly and records successful transactions.
- To ensure that admin CRUD operations correctly update records in MongoDB.
- To confirm that unauthorized visitors cannot access protected admin routes.

6.3 Levels of Testing

The testing strategy employed for this project comprises four standard levels:

- **1. Unit Testing:** Individual modules, utility functions (e.g. CAPTCHA verification, password hashing, JWT token generation), and isolated React components were tested independently.
- **2. Integration Testing:** Verified data flow between React frontend components and Express backend API routes, ensuring JSON payloads are received and processed correctly.
- **3. System Testing:** The entire application was tested end-to-end (from user registration, shoe search, and cart addition to checkout and admin order status update) to ensure seamless workflow.
- **4. Acceptance / Usability Testing:** Conducted with peers and group members to verify that navigation is simple, text is readable, and buttons respond promptly.

6.4 Test Cases

The table below documents 12 actual test cases executed on the Online Shoes Website:

TC ID	Feature Tested	Input / Action	Expected Result	Actual Result	Status
TC-01	User Registration	Enter valid name, new email, password, address, and correct CAPTCHA	Account created, user logged in, redirected to home page	Account created, JWT stored in localStorage	PASS
TC-02	Duplicate Email Check	Register with an already existing email address	Error message: "User with this email already exists"	Displayed error alert, registration prevented	PASS
TC-03	Customer Login	Enter valid email (rahul@gmail.com), password (user123), and valid CAPTCHA	Successful login, customer name displayed in navbar	Logged in, redirected to store	PASS
TC-04	Invalid Password Login	Enter correct email with wrong password	Error message: "Invalid email or password"	Displayed 401 error message	PASS
TC-05	Shoe Search	Type "Nike" in navbar search bar and press Enter	Products page displays only Nike brand shoes	Filtered list showing Nike shoe models	PASS
TC-06	Category Filtering	Click on "Running" category badge	Display only shoes categorized under Running	Catalog updated to show running shoes	PASS
TC-07	Add to Cart with Size	Select UK Size 8 on product details and click "Add to Cart"	Shoe added to cart with Size UK 8, cart count badge increments	Cart badge updated to 1, item saved in CartContext	PASS
TC-08	Cart Quantity Update	Click '+' button to increase shoe quantity to 2	Subtotal doubles, total payable updates accurately	Amount updated immediately in cart table	PASS
TC-09	Checkout Validation	Leave shipping address empty and click Pay button	Browser HTML5 validation prompts user to fill required fields	Submission stopped, field highlighted	PASS
TC-10	Razorpay Payment	Click "Pay with Razorpay" with valid delivery address	Official Razorpay modal popup opens with payment options	Razorpay checkout window	PASS

TC ID	Feature Tested	Input / Action	Expected Result	Actual Result	Status
				opened successfully	
TC-11	Admin Portal Access	Login at /admin/login with admin@shoestore.com / admin123	Access granted to Admin Dashboard, client navbar hidden	Admin dashboard loaded with sidebar navigation	PASS
TC-12	Admin Order Status	Change order status from "Pending" to "Delivered"	Status updated in MongoDB and badge changes to green Delivered	Status updated and badge displayed as Delivered	PASS

7 LIMITATION AND SYSTEM ENHANCEMENT

7.1 Limitation

Although the Online Shoes Website successfully fulfills all primary requirements of an online footwear store, it has a few practical limitations typical of a student academic project:

- **Single Currency Support:** The system currently transactions solely in Indian Rupees (INR — ₹). International multi-currency conversion is not supported.
- **Single Vendor Store:** The application is designed for a single shoe store owner/business. It does not support a multi-vendor marketplace model where third-party merchants can register and sell their own footwear.
- **Email Notification Service:** Order receipts and status updates are displayed on screen and stored in the database, but automated email notifications (via SMTP / Nodemailer) are not yet integrated.
- **Size Recommendation:** The website provides standard UK shoe sizes (6 to 11) but lacks an interactive foot measurement or 3D sizing guide tool.
- **Internet Dependency:** Certain third-party assets such as Google Fonts and the Razorpay checkout script require an active internet connection to load.

7.2 System Enhancement

To further expand the system for future commercial use, the following enhancements are planned:

- **Automated Email & SMS Alerts:** Integrating Nodemailer and SMS APIs (such as Twilio) to dispatch automatic order confirmations, shipping tracking numbers, and delivery alerts to customers.
- **Interactive Shoe Size Advisor:** Adding an interactive foot length measurement calculator to help customers select the ideal shoe size and reduce size-exchange rates.
- **Customer Returns & Refund Portal:** Introducing a dedicated module allowing customers to initiate return requests and administrators to approve refunds.
- **Coupon & Discount Code Engine:** Enabling store managers to create promotional discount promo codes with percentage discounts and expiry dates.
- **Progressive Web App (PWA) Support:** Enabling offline catalog caching and installable mobile app capabilities on Android and iOS devices.

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The development of this project and preparation of documentation were supported by the following official documentation, textbooks, and web references:

Official Documentation & Online References

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3. **Express.js API Reference:** <https://expressjs.com/> — Reference for RESTful API routing, middleware, and request/response handling.
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5. **Razorpay Developers Documentation:** <https://razorpay.com/docs/> — Reference for standard checkout modal integration and payment signature verification.
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